



Reviewing the carbonation resistance of low-clinker Limestone Calcined Clay Cements (LC³) - Future directions using non-destructive imaging methods

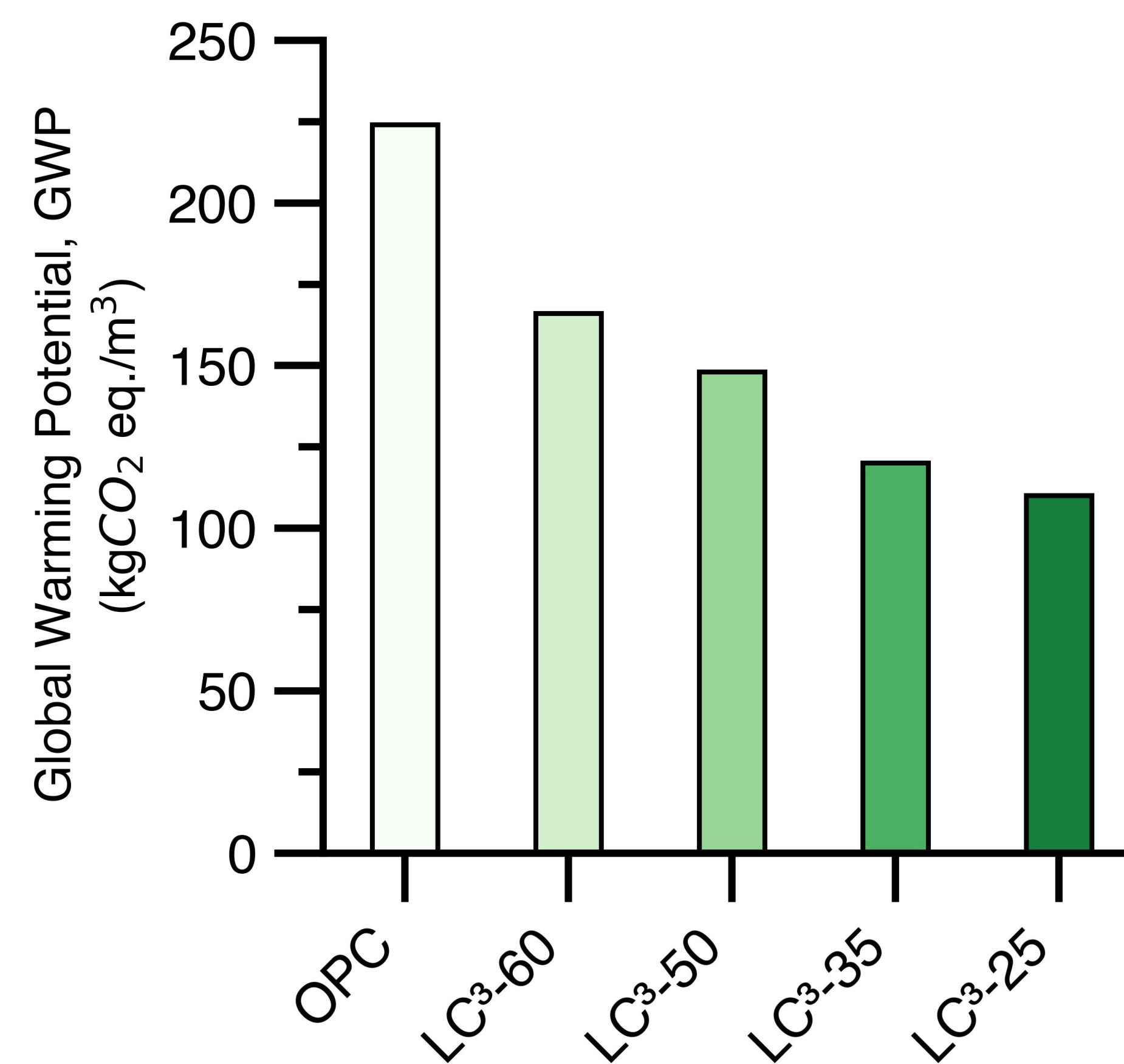
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Introduction and Research Significance

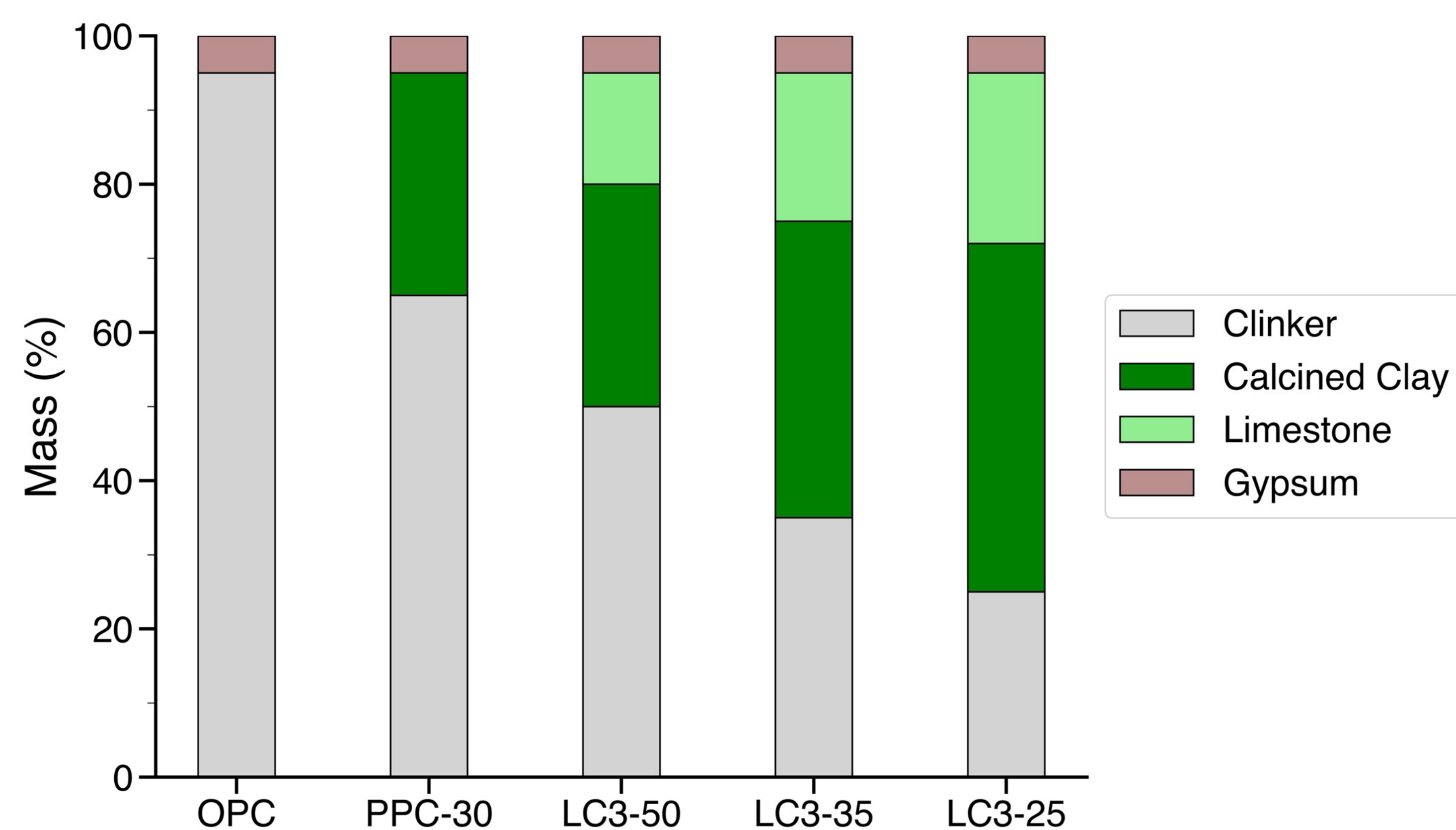
Corrosion of steel in concrete is the most significant cause of concrete deterioration. Current design doctrines believe reduced alkalinity in low-clinker concrete increases the possibility of steel corrosion. This philosophy stymies the use of such binder systems in reinforced concrete. The buffer capacity of these systems further reduces as we move towards <40% clinker. However, the possibility of improving this buffer capacity in such systems is yet to be explored. This study reviews the buffer capacity of low-clinker systems, highlighting factors influencing carbonation resistance. To evaluate the carbonation resistance, proof of concept non-destructive X-Ray Computed Tomography (X-CT) experiments were conducted. Such studies are vital and provide a paradigm shift of carbonation-induced steel corrosion in such low-clinker systems via non-destructive test methods.

Global Warming Potential of Low-Clinker Blends



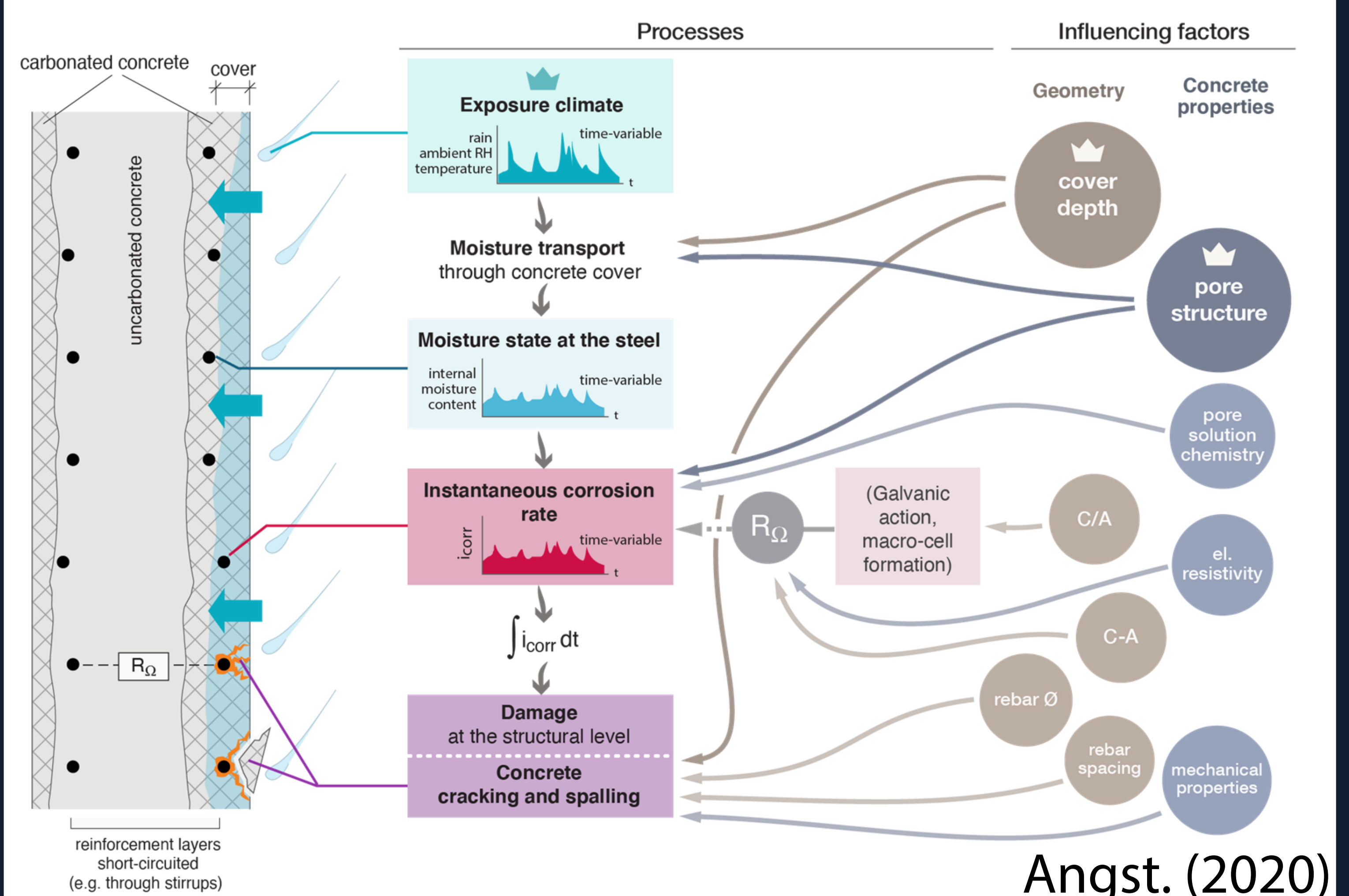
Reduction in Global Warming Potential, evident upon reduction in clinker. However, dilution also occurs where a certain minimum amount of clinker is required for optimum sustainability.

Limestone Calcined Clay Cement



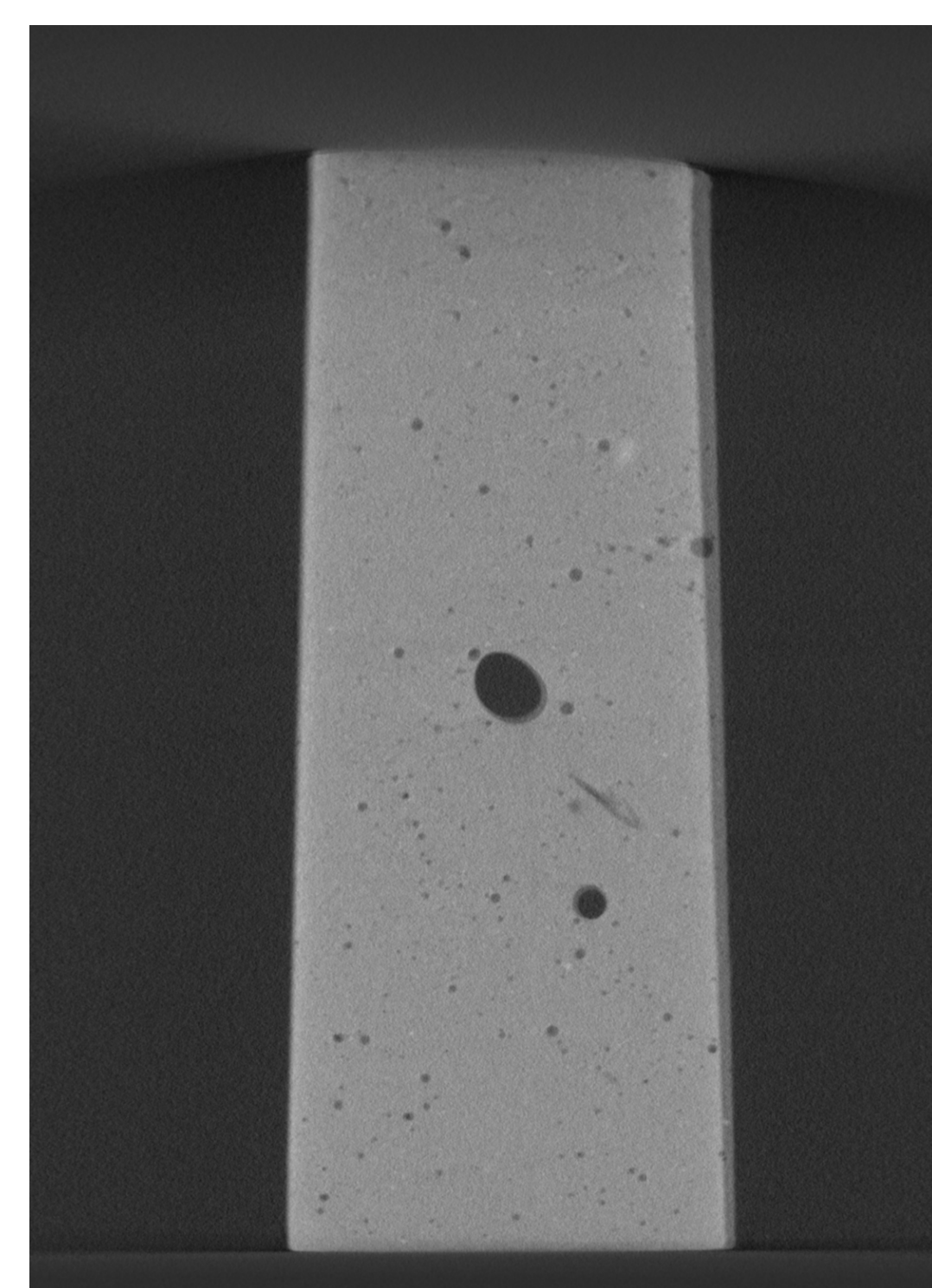
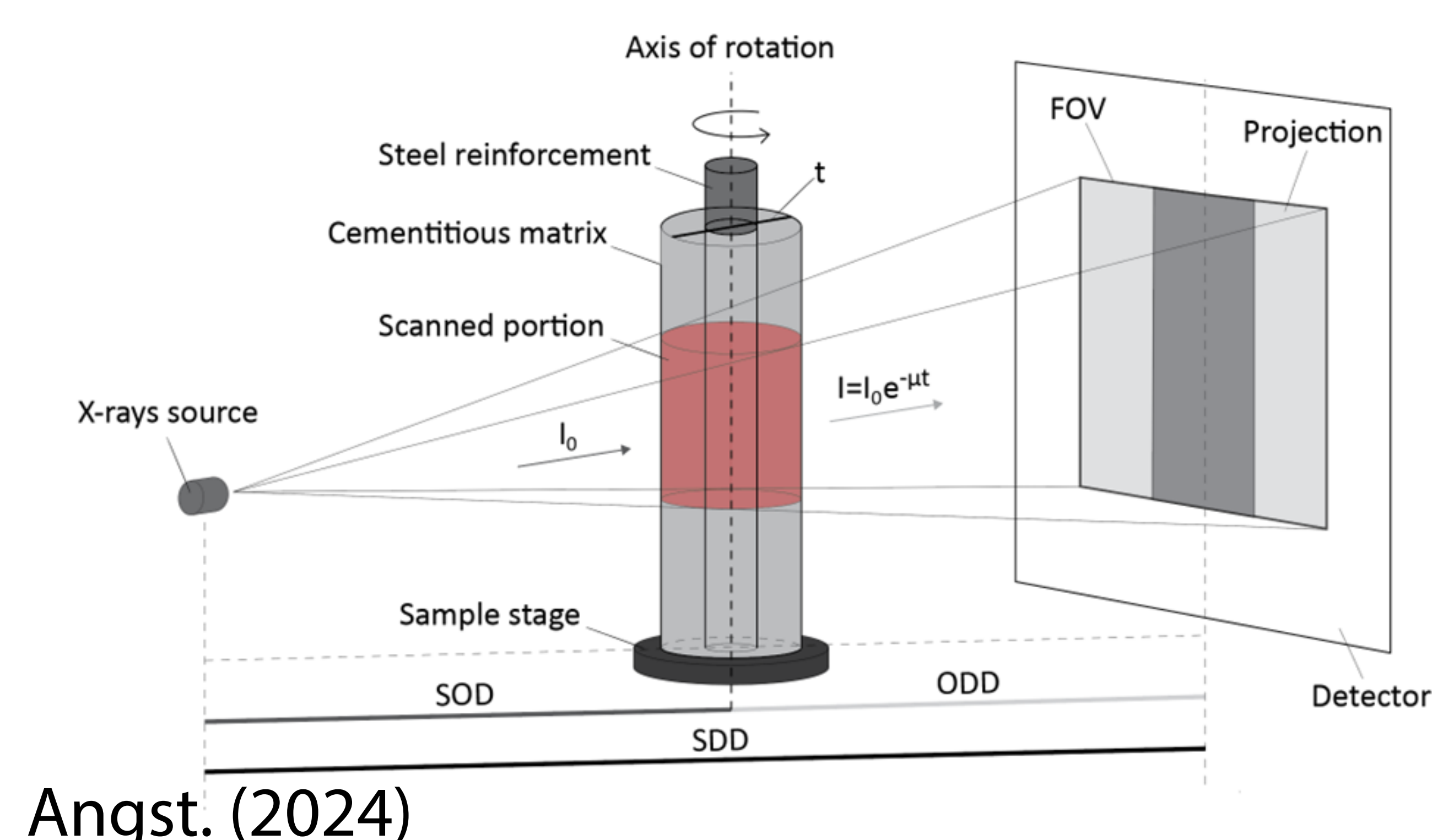
Limestone Calcined Clay Cement (LC³) substitutes clinker or cement with clay and limestone. The availability of clay and limestone in regions where construction demand is rising offers LC3's feasible integration into construction.

Carbonation Performance



Several factors influence the carbonation resistance of concrete. Low-clinker and its reduced buffer, might not necessarily lead to steel corrosion.

Non-Destructive Evaluation of the Carbonation Affected Zone via X-Ray Computed Tomography



CO₂ ingress Using preliminary X-CT images, we are able to image the carbonation front. The next steps would be to image the carbonation affected zone with steel-reinforcement in place, to understand corrosion damage.

Acknowledgements



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References

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2. Rathnakumar, S., Garg, N. *Manuscript in preparation.*